

Plant Inspection Report

Refinery Unit 4 - Annual Mechanical Integrity Inspection

Facility: Example Refinery, Block 4 Report number: IR-2026-0417 Inspection window: 06 April 2026 to 10 April 2026

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1. Scope

This report covers the static and rotating equipment in Unit 4, the associated firewater system, the electrical distribution room MCC-7 and the cooling water circuit. The inspection used visual examination, thermographic survey of electrical panels and pipework, and thickness measurement at designated inspection points.

2. Overall assessment

The unit is fit for continued operation. Three findings are classified HIGH and require corrective action within thirty days. No immediate shutdown condition was identified. The overall risk rating for Unit 4 is MEDIUM.

3. Findings summary

ID	Severity	Location	Finding	Recommendation	Due
F-001	HIGH	Distillation Column DC-2	Tray corrosion on stage 7, estimated 15% metal loss	Replace tray set during next shutdown	2026-11-04
F-002	MEDIUM	Firewater Pump P-301B	Diesel day tank level sensor reads 4% high	Recalibrate LT-301B	2026-10-20
F-003	HIGH	Heat Exchanger E-204	Gasket weep observed at channel 3 flange	Replace gasket within 30 days	2026-10-15
F-004	LOW	Electrical MCC-7	Panel lighting failure, two lamps not working	Replace lamps at next opportunity	2027-01-31
F-005	MEDIUM	Cooling Tower CT-1	Approach 3.2 C against a limit of 3.0 C	Inspect and clean cooling media	2026-10-28

4. Detailed findings

F-001 Distillation Column DC-2 tray corrosion (HIGH)

Ultrasonic thickness readings on stage 7 trays showed a minimum remaining thickness of 11.2 mm against a design thickness of 13.2 mm, representing approximately fifteen percent metal loss. Corrosion was localised on the east side of the tray deck adjacent to the feed entry nozzle. The tray support bracket showed measurable pitting. Recommendation is to replace the tray set and inspect the bracket during the next planned shutdown, which is scheduled for 04 November 2026.

F-002 Firewater Pump P-301B day tank level transmitter (MEDIUM)

During functional testing the day tank level transmitter LT-301B indicated a level of four percent higher than the actual gauged level. The firewater pump start test passed, so the deficiency is one of indication accuracy rather than delivery capacity. Recalibration is required before the next fire drill.

F-003 Heat exchanger E-204 channel 3 flange gasket (MEDIUM)

A small weep was observed at the channel three inlet flange during the thermographic survey at ambient conditions. The flange temperature differential was 8 C above the adjacent flange, confirming an active leak path. The gasket is scheduled for replacement within thirty days as a planned defect.

5. Detailed findings continued

F-004 Electrical MCC-7 panel lighting (LOW)

Two of the six panel lamps in MCC-7 failed. This is an ergonomic finding that does not affect protection of the busbar. Lamp replacement is deferred to the next available maintenance window.

F-005 Cooling Tower CT-1 approach temperature (MEDIUM)

The cooling water approach temperature measured 3.2 C against a design limit of 3.0 C. Drift loss was not identified as the cause. Inspection of the cooling media and a basin water quality analysis is recommended.

6. Conclusion and recommendation

Unit 4 may continue to operate. The three HIGH findings F-001, F-002 and F-003 must be closed by their due dates. A follow-up verification inspection is recommended within sixty days of the last corrective action.

7. Signatures

Lead inspector: R. Sharma, QA/QC Reviewed by: T. Almeida, Head of Integrity Approved by: K. Osei, Plant Manager